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MARMARA UNIVERSITY
FACULTY OF ENGINEERING

EE4002
Digital Signal Processing

Lab 4

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Objective

The objective of this laboratory session is to introduce students to basic STM32 DAC usage for signal generation and familiarize them with waveform visualization using Analog Discovery 2. Students will generate simple sinusoidal and cosine signals and explore the effects of superposition by combining them into a composite signal.

Students will:

- Generate specific sine and cosine signals with given amplitudes and frequencies using STM32.
- Visualize and verify the generated signals using Analog Discovery 2.
- Combine multiple signals using superposition and analyze the resulting waveform.

Equipment

- STM32 Nucleo/Discovery Board
- STM32CubeIDE
- Analog Discovery 2
- Digilent WaveForms Software

Pre-Lab Assignment

Purpose:

The objective of this laboratory session is to introduce students to basic STM32 DAC usage for signal generation and to familiarize them with waveform visualization using Analog Discovery 2. Students will generate simple sinusoidal and cosine signals with specified parameters, and explore the effects of superposition by combining them into a composite signal.

Tasks:

1. Before attending the laboratory session, students are expected to simulate the intended waveforms in MATLAB. This preliminary step will allow them to gain familiarity with the expected signal shapes and verify that the waveform parameters (frequency, amplitude, phase) are correctly interpreted.

2. Please review the connections I have added before the lab session. I am sharing two links below should check beforehand.https://wiki.st.com/stm32mcu/wiki/Getting_started_with_DAC , https://www.youtube.com/watch?v=JBbT_3wgzx8

Procedure

Procedure 1: Individual Signal Generation and Visualization

Procedure 1A: 15 Hz Sine Wave, Amplitude = 1

- Use STM32 DAC to generate a sine wave with a frequency of 15 Hz and an amplitude of 1.
- Observe the waveform using Analog Discovery 2.

Procedure 1B: 30 Hz Sine Wave, Amplitude = 0.5

- Use STM32 DAC to generate a sine wave with a frequency of 30 Hz and an amplitude of 0.5.
- Observe the waveform using Analog Discovery 2.

Procedure 1C: 25 Hz Cosine Wave, Amplitude = 0.8

- Use STM32 DAC to generate a cosine wave with a frequency of 25 Hz and an amplitude of 0.8.
- Observe the waveform using Analog Discovery 2.

Procedure 2: Superposition of Signals

- Using STM32, combine the three previously generated signals into a single signal using superposition.
- Output the combined waveform through DAC.
- Use Analog Discovery 2 to visualize and analyze the resulting waveform.

Discussion

- How accurately were you able to generate the specified frequencies and amplitudes?
- What happens when multiple sinusoids are combined? Describe the shape of the resulting waveform.
- How does the use of different amplitudes affect the visibility of individual components in the combined signal?
- What are some real-world applications of signal superposition?

Lab Questions

1. What are the key parameters that define a sine or cosine wave?
2. How can you ensure precise frequency and amplitude when using STM32 DAC?
3. What is signal superposition, and how is it implemented in this experiment?

Deliverables

- Include screenshots of the waveforms in your report (provide explanations in the image captions).
- Write a brief report answering the provided questions.
- Submit waveform comparisons, screenshots, and a short analysis.
- Upload the final report as a **single PDF file (Canvas submission).**